USE CASE DIAGRAM

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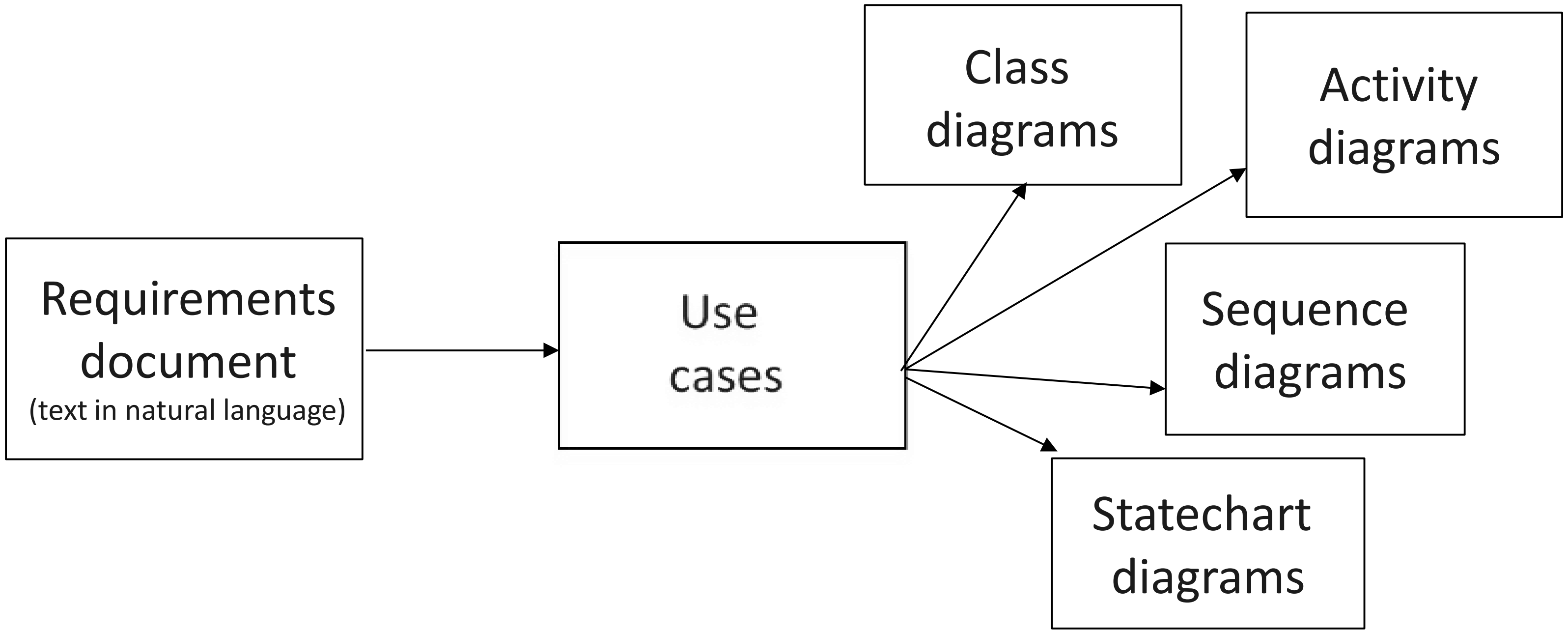
WHAT IS A USE CASE DIAGRAM

- A Use Case Diagram is a graphical representation of the functional requirements of a system.
- It shows how users interact with the system and the actions they take to achieve their goals.
- Use Case Diagrams are part of the broader context of system analysis, which aims to identify, describe, and organize the requirements of a system.

USE CASE DIAGRAM (CONTINUATION)



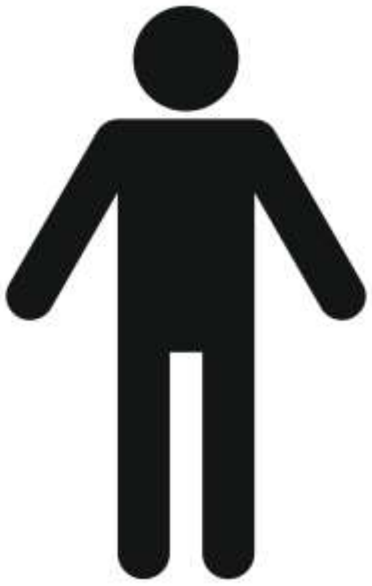
Use case models are developed at different levels of abstraction, and it
Is an iterative and incremental process.

COMPONENTS OF A USE CASE DIAGRAM

- Actors
- Use cases
- System Boundary
- Relationship

COMPONENTS OF A USE CASE DIAGRAM (CONTINUATION)

- **Actors** are the people, organizations, or systems that interact with the system being modeled.
- They can be internal or external to the system and can have different roles and levels of access.
- Examples of actors in a Use Case Diagram are customers, administrators, and third-party services or systems.



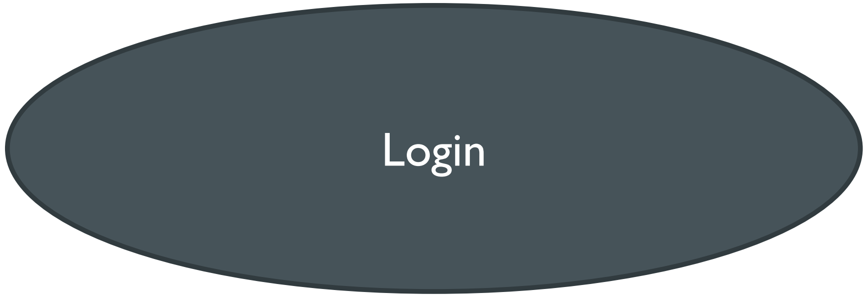
Actors are represented by a stick figure

TYPES OF ACTORS

- **Primary Actors**
  - These are the actors that directly interact with the system to achieve their goals or complete their tasks.
  - They can be human, System or Software, Hardware, and Timer or Clock.
- **Secondary Actors**
  - These are the actors that indirectly interact with the system to support or assist the primary actors in achieving their goals.
  - Examples of secondary actors are Third Party Payment Gateways, and External Databases

COMPONENTS OF A USE CASE DIAGRAM (CONTINUATION)

- A **Use Case** is a specific set of actions that a user takes to accomplish a goal or task within the system.
- There are different types of Use Cases.
- Examples of Use Cases in a Use Case Diagram are login, checkout, and search.



A Use Case is represented by an oval

TYPES OF USE CASES

- **Essential Use Cases**: These are the use cases that represent the core functionality of the system. They describe the most important goals or tasks that the system is designed to accomplish.
- **Supporting Use Cases**: These are the use cases that support the essential use cases. They describe the additional functionality that is necessary to achieve the essential goals or tasks.
- **User-Level Use Cases**: These use cases represent the goals or tasks that are performed by the end-users of the system. They describe the functionality that is visible to the user.
- **System-Level Use Cases**: These use cases represent the goals or tasks that are performed by the system itself. They describe the internal functionality of the system that is not visible to the user.

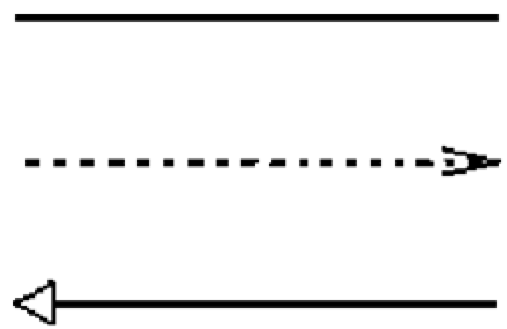
COMPONENTS OF A USE CASE DIAGRAM (CONTINUATION)

- A **System Boundary** separates the use cases that are internal to the system from the actors that are external to the system.

A System Boundary is represented by a rectangle

COMPONENTS OF A USE CASE DIAGRAM (CONTINUATION)

- **Relationships** are the association between actors and use cases.

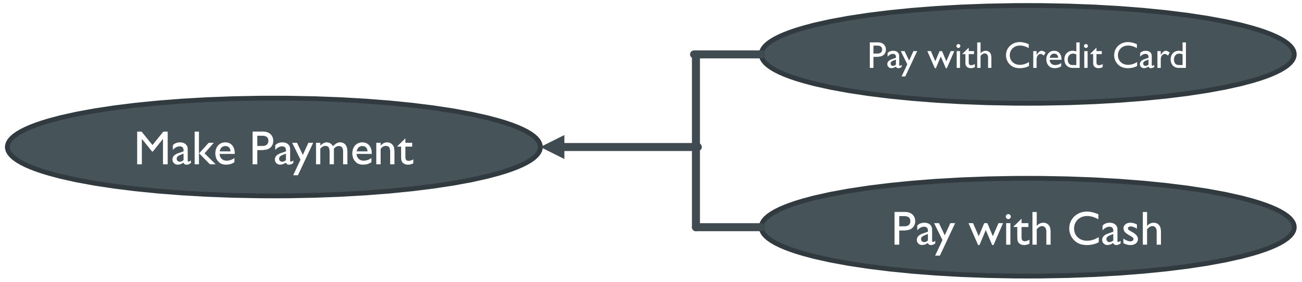


Depending upon the type of relationship, several symbols are used.

TYPES OF RELATIONSHIPS (CONTINUATION)

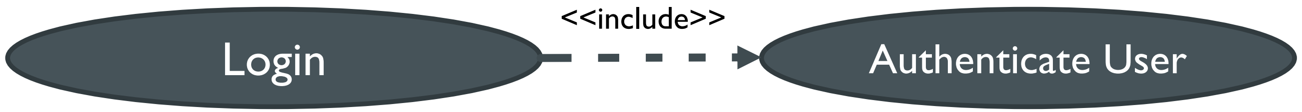
- **Association**: An association relationship between two use cases indicates that they are "related" in some way. This relationship is represented by a solid line connecting the two use cases. For example, a use case for "Make Payment" might be associated with a use case for "View Cart."

- **Generalization**: A generalization relationship between two use cases indicates that one use case is a "general" version of the other. This relationship is represented by an arrow pointing from the more specific use case to the more general use case. For example, a use case for "Pay with Credit Card" and "Pay with Cash" might be a specialization of a more general use case for "Make Payment."

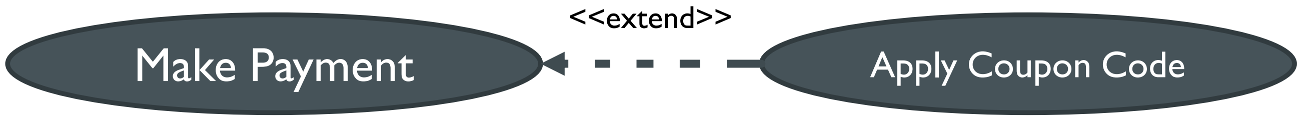


TYPES OF RELATIONSHIPS

- **Include**: An include relationship between two use cases indicates that one use case is a part of another use case. It means that the included use case is "always" performed when the including use case is executed. This relationship is represented by a dashed arrow pointing from the including use case to the included use case labeled with <<include>>. For example, a use case for "Authenticate User" might be included in the "Login" use case.



- **Extend**: An extend relationship between two use cases indicates "optional" use of the functionality of another use case. This relationship is represented by a dashed arrow pointing from the extending use case to the extended use case labeled with <<extend>>. For example, a use case for "Apply Coupon Code" might be extend from the "Make Payment" use case.



USE CASE EXAMPLE SCENARIO

- A patient calls the clinic to make an appointment for a yearly checkup. The management finds the nearest empty time slot in the appointment book and schedules the appointment for that time slot where the doctor records his availability. "

USE CASE EXAMPLE SCENARIO (IDENTIFY THE ACTORS)

- A patient calls the clinic to make an appointment for a yearly checkup. The management finds the nearest empty time slot in the appointment book and schedules the appointment for that time slot where the doctor records his availability. "

- Actors: Patient, Management, Doctor

USE CASE EXAMPLE SCENARIO (IDENTIFY THE USE CASES)

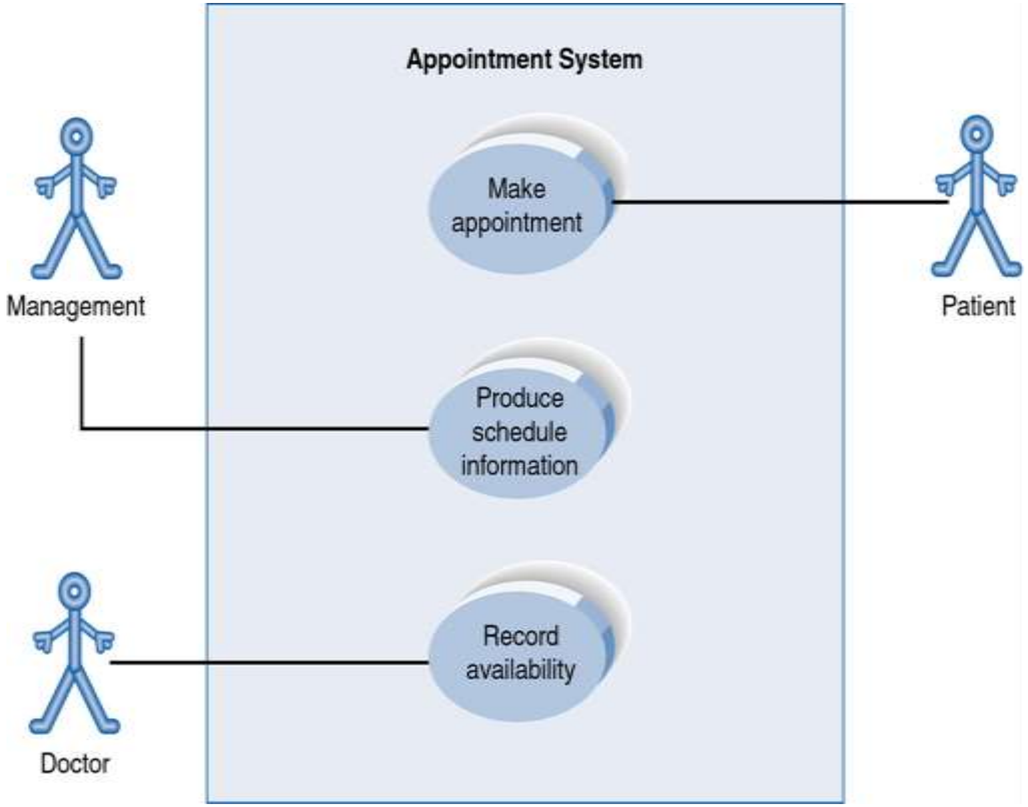
- A patient calls the clinic to make an appointment for a yearly checkup. The management finds the nearest empty time slot in the appointment book and schedules the appointment for that time slot where the doctor records his availability. "

- Actors: Patient, Management, Doctor
- Use Cases: Make an Appointment, Finds the nearest empty time slot in the appointment book and schedules the appointment for that time slot, Record his availability

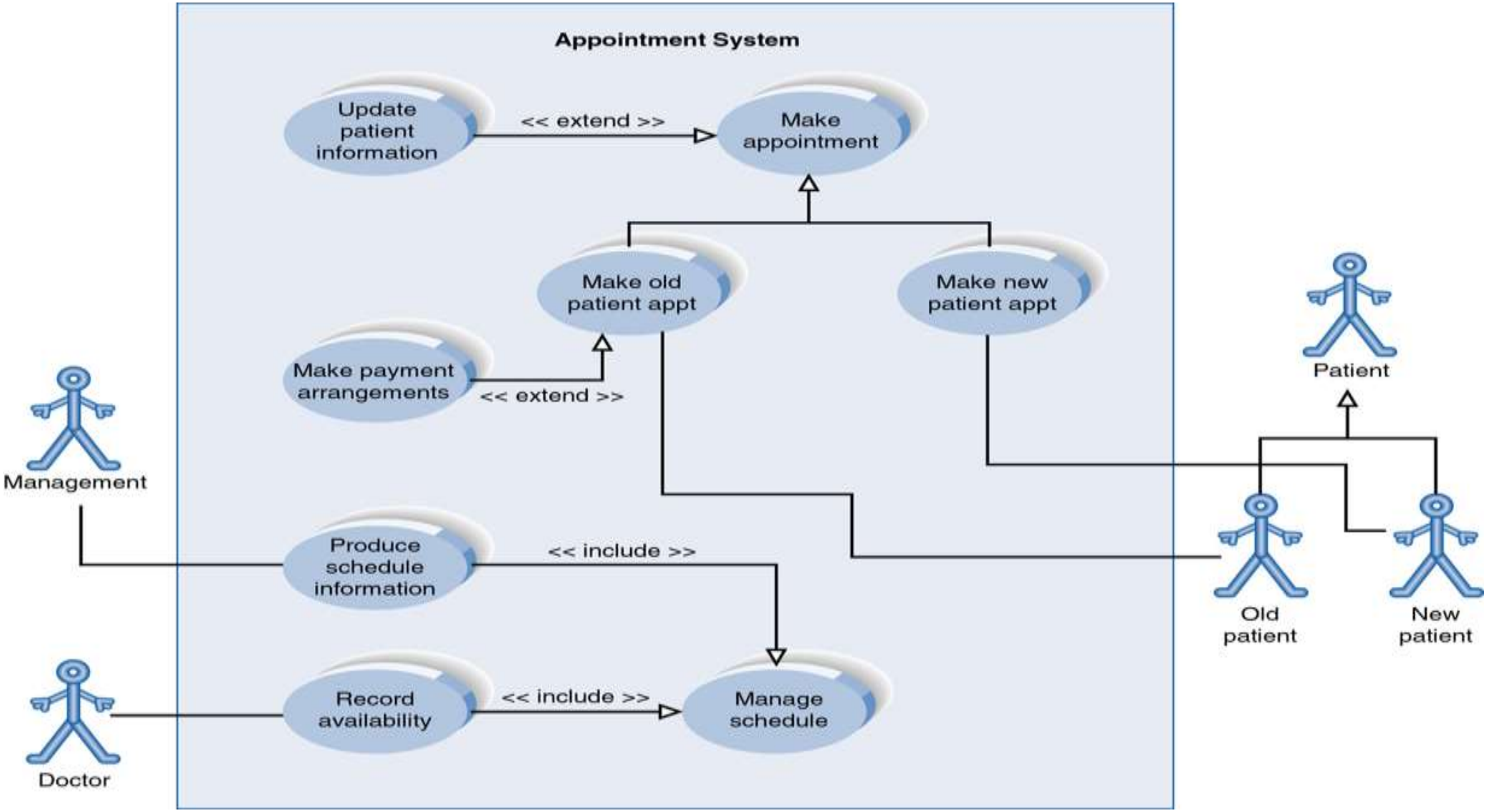
USE CASE EXAMPLE SCENARIO (COMPLETE THE DIAGRAM)

- A patient calls the clinic to make an appointment for a yearly checkup. The management finds the nearest empty time slot in the appointment book and schedules the appointment for that time slot where the doctor records his availability. "

- Actors: Patient, Management, Doctor
- Use Cases: Make Appointment, Produce Schedule Information, Record Availability



ANOTHER EXAMPLE



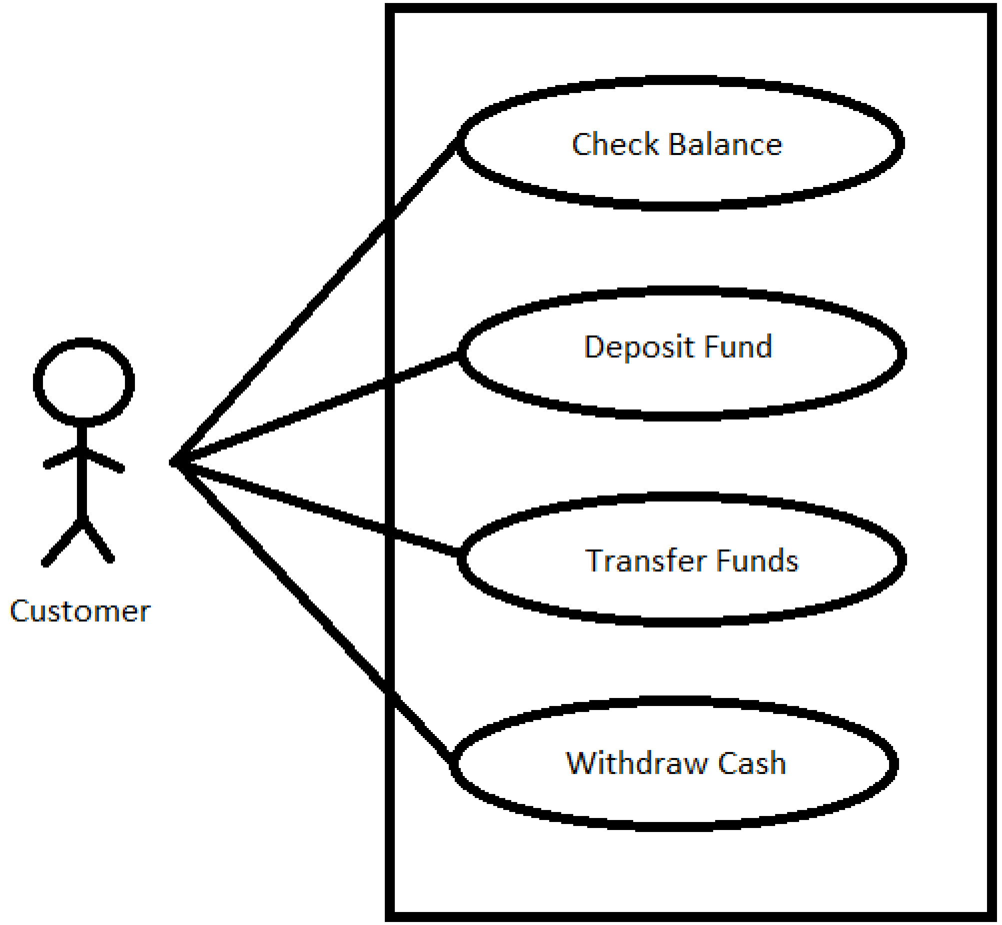
USE CASE DIAGRAM ACTIVITY

An automated teller machine is a banking subsystem that provides bank customers with access to financial transactions in a public space without the need for a cashier, clerk, or bank teller.

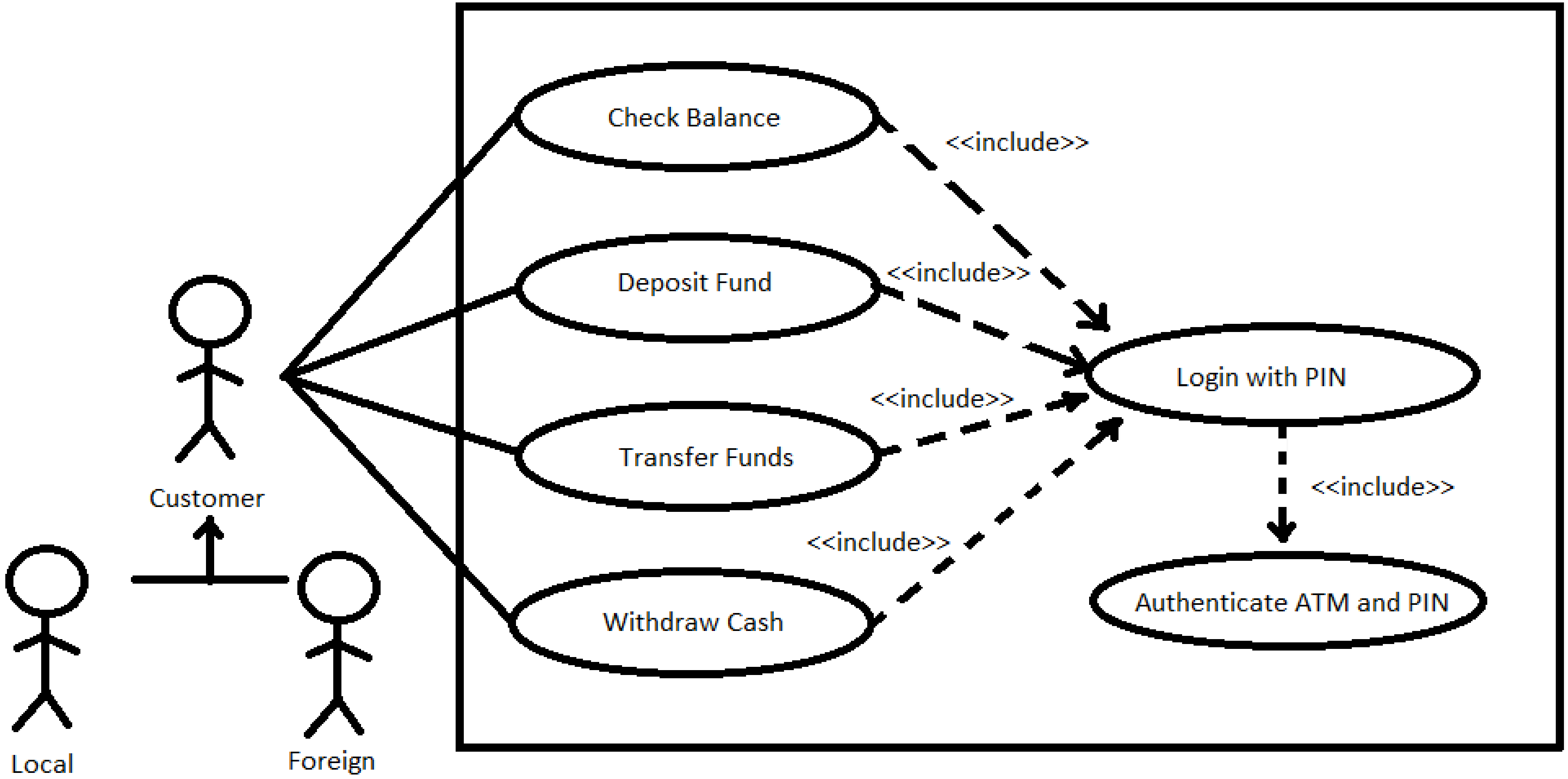
Scenario:

A Customer, which can either be a local or foreign, uses an ATM to Check Balances, Deposit Funds, Transfer Funds, and Withdraw Cash all after his account is verified thru the ATM Card and pin. Design the Use Case Diagram.

ANSWER USE CASE DIAGRAM (SIMPLE)



ANSWER USE CASE DIAGRAM (WITH RELATIONSHIPS)



USE CASE DIAGRAM ACTIVITY (ADDITIONAL SCENARIO)

An automated teller machine is a banking subsystem that provides bank customers with access to financial transactions in a public space without the need for a cashier, clerk, or bank teller.

Scenario:

A Customer, which can either be a local or foreign, uses an ATM to Check Balances, Deposit Funds, Transfer Funds, and Withdraw Cash all after his account is verified thru the ATM Card and pin. Design the Use Case Diagram.

Additional Scenario:
The ATM will ask the Customer if he wants to have a printed receipt or not after withdrawing cash.

ANSWER USE CASE DIAGRAM WITH THE ADDITIONAL SCENARIO